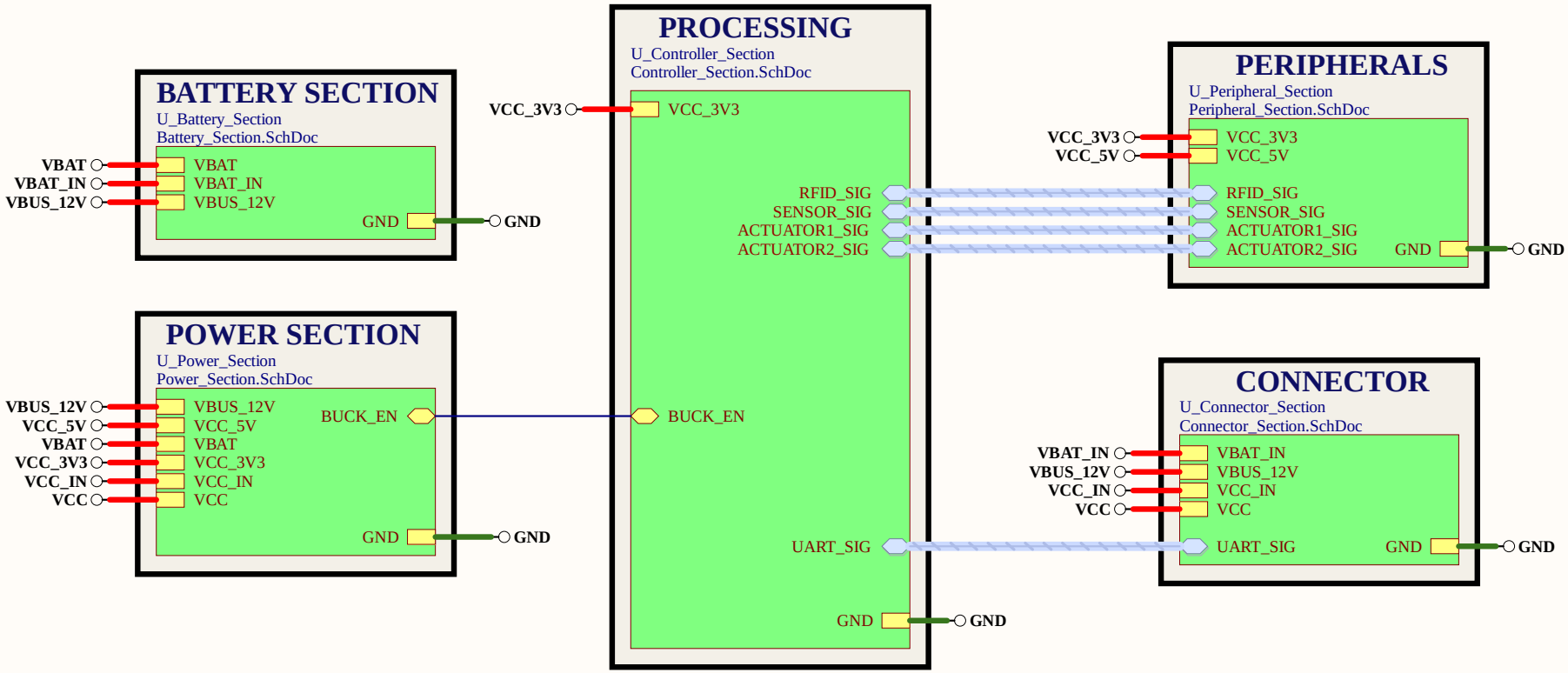
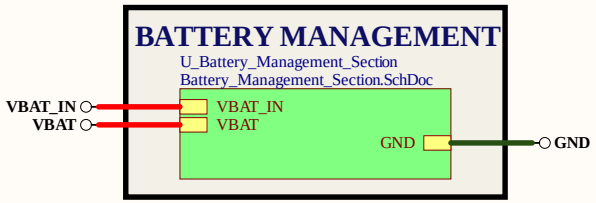
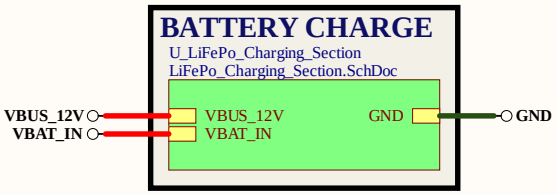
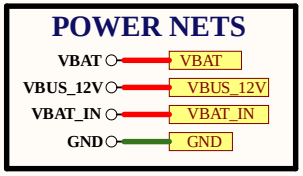
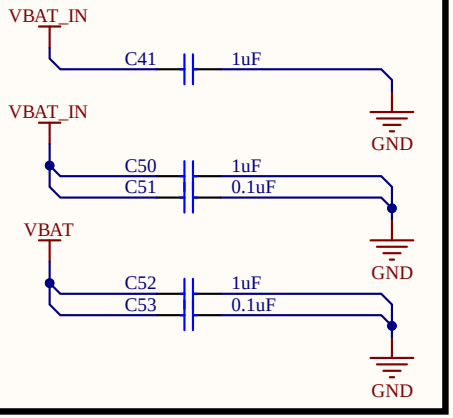

BATTERY SECTION
U_Battery_Section
Battery_Section.SchDoc
VBAT
VBAT_IN
VBUS_12V
GND
VBAT
VBAT_IN
VBUS_12V
GND
POWER SECTION
U_Power_Section
Power_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
PROCESSING
U_Controller_Section
Controller_Section.SchDoc
VCC_3V3
VCC_3V3
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
BUCK_EN
UART_SIG
GND
GND
PERIPHERALS
U_Peripheral_Section
Peripheral_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
GND
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
GND
CONNECTOR
U_Connector_Section
Connector_Section.SchDoc
VBAT_IN
VBUS_12V
VCC_IN
VCC
VBAT_IN
VBUS_12V
VCC_IN
VCC
UART_SIG
GND
UART_SIG
GND


BATTERY MANAGEMENT
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT_IN
VBAT
VBAT_IN
VBAT
GND
GND

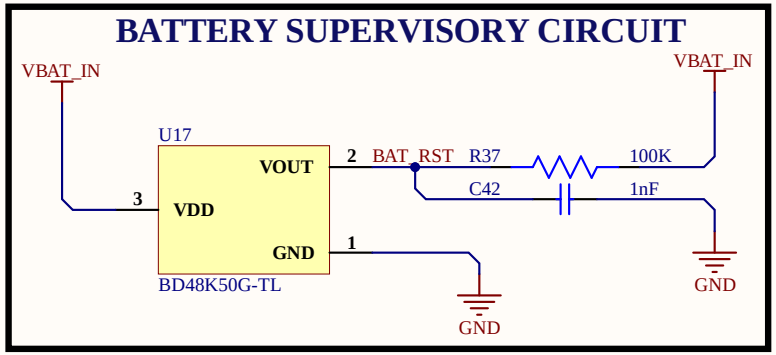

BATTERY CHARGE
U_LiFePo_Charging_Section
LiFePo_Charging_Section.SchDoc
VBUS_12V
VBAT_IN
VBUS_12V
VBAT_IN
GND
GND


POWER NETS
VBAT
VBUS_12V
VBAT_IN
GND
VBAT
VBUS_12V
VBAT_IN
GND

DECOUPLING CAPACITOR

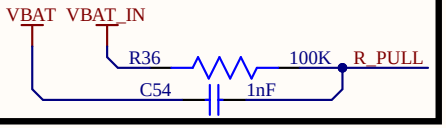


BATTERY SUPERVISORY CIRCUIT

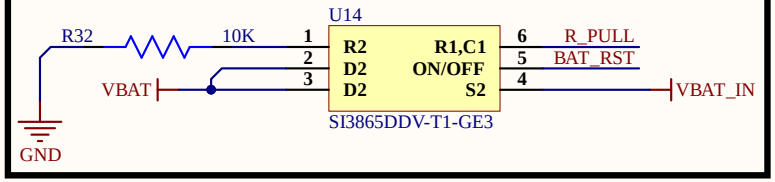


Detection voltage: 2.3V to 6.0V (Typ.)
Voltage threshold: 5V

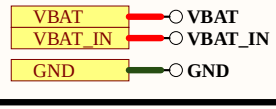
PULL UP RESISTOR



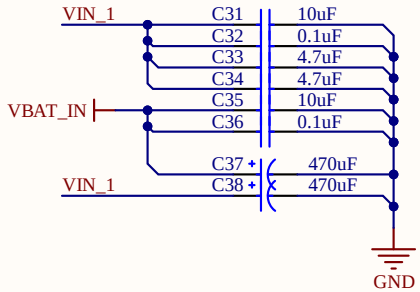
POWER SWITCH IC



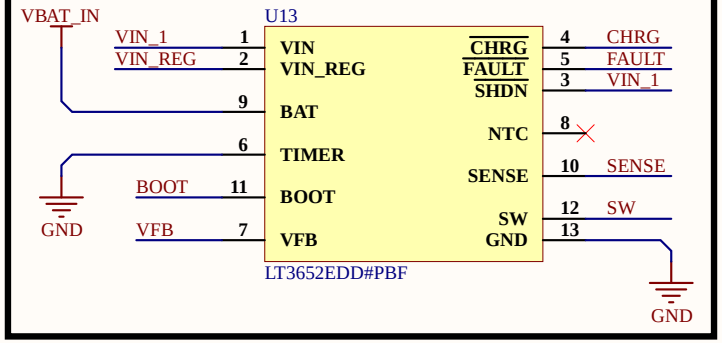
POWER NETS



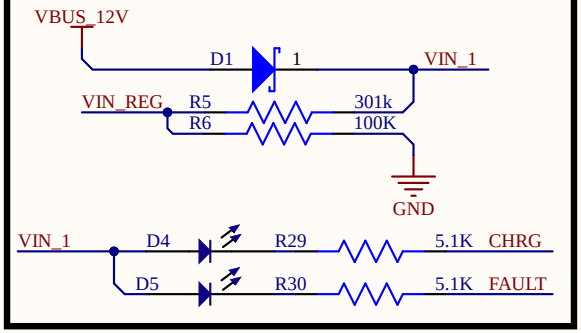
DECOUPLING CAPACITORS



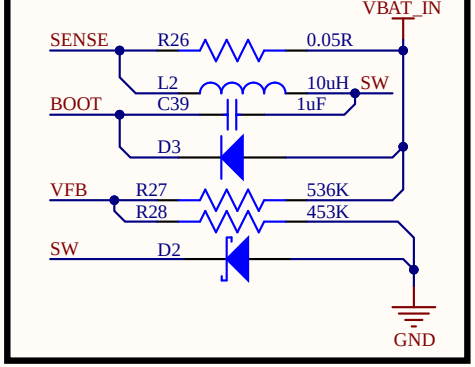
BATTERY CHARGING CIRCUIT



SUPPORTING CIRCUITRY

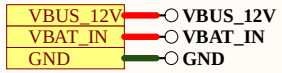


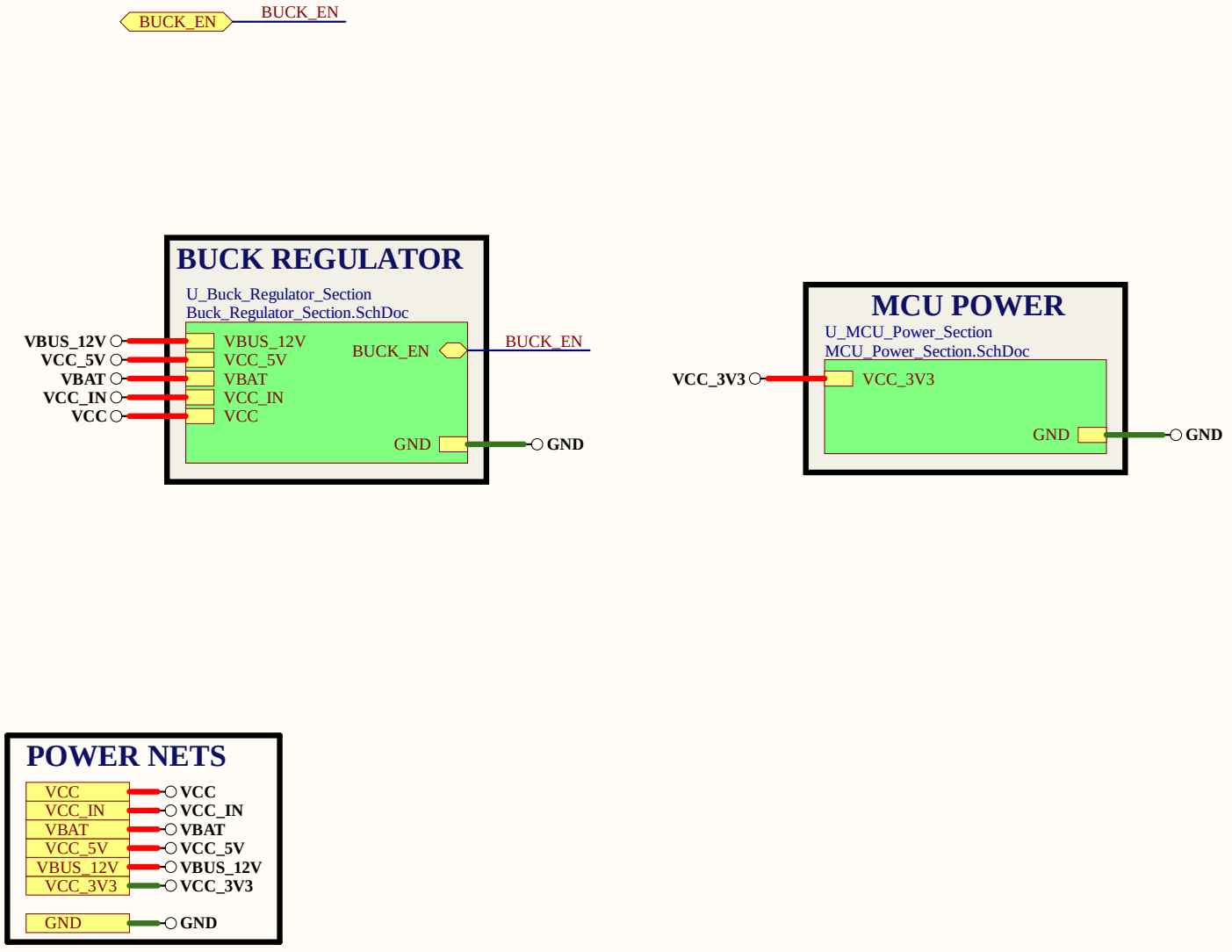
BATTERY INPUT CIRCUIT



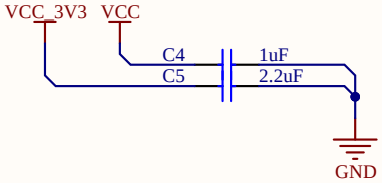
Battery Chemistry: 2 Cell LiFeOP4
Input Voltage: 12V USB
Battery Voltage: (2 × 3.6V) = 7.2V
Charge current: 2A

POWER NETS

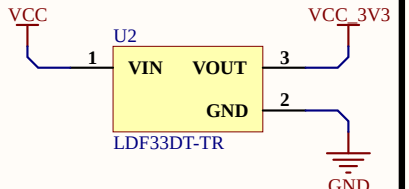



BUCK_EN
BUCK_EN
BUCK REGULATOR
U_Buck_Regulator_Section
Buck_Regulator_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
BUCK_EN
GND
BUCK_EN
GND
MCU POWER
U_MCU_Power_Section
MCU_Power_Section.SchDoc
VCC_3V3
VCC_3V3
GND
GND
POWER NETS
VCC
VCC_IN
VBAT
VCC_5V
VBUS_12V
VCC_3V3
GND
VCC
VCC_IN
VBAT
VCC_5V
VBUS_12V
VCC_3V3
GND

DECOUPLING CAPACITOR



3V3 REGULATOR



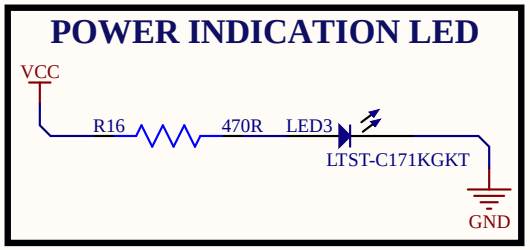
Input Voltage: 12V USB or 7.2V VBAT
Output Current: 1A
Output Voltage: Fixed 3.3V
Voltage Drop: 500 mV max. at 1 A load

- Calculator value for current limiting Resistance: 4 8 0 R for 20mA @VCC=12V

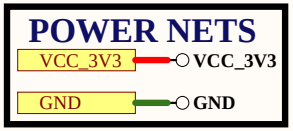
- Closest standard value=470R

LED max current rating=30mA (Datasheet =pg:3)
LED test current= 20mA (pg:5)

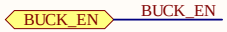
POWER INDICATION LED



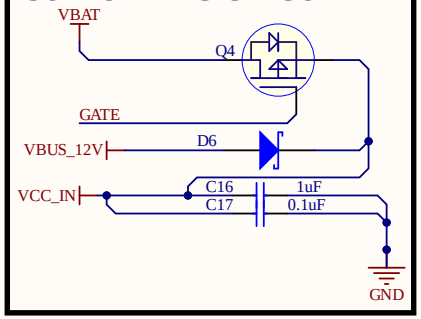
POWER NETS



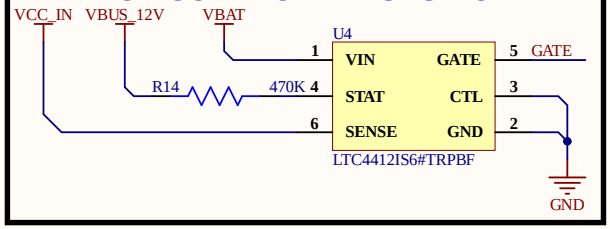
BUCK ENABLE



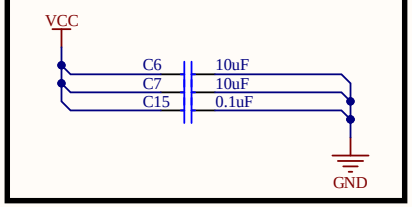
SUPPORTING CIRCUITRY



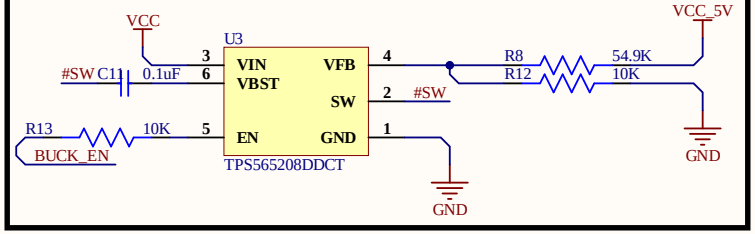
OR CONTROLLER SECTION



DECOUPLING CAPACITOR



BUCK SWITCHING REGULATOR CIRCUIT



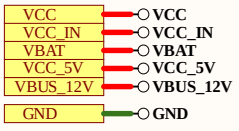
3.3uH inductor and output decoupling capacitor missing in the current design. (pg: 1,14.... datasheet)

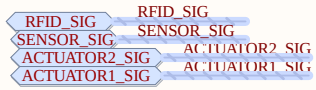
- Input Voltage: switches between 12V USB and VBAT (7.2V)
- Output Voltage: 5V
- Output Current: 5A Maximum.

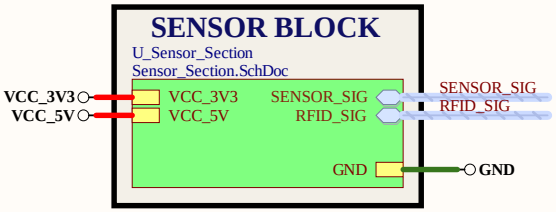
Output Voltage Calculation:

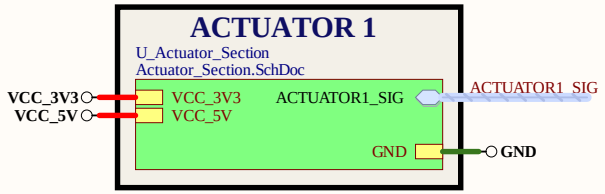
Formula: V =0.760 (1+R1/R2) (pg:13)
R1= 54.9K, R2=10K, L=3.3uH (pg:14)
VOUT= 4.93V
UVLO threshold: 3.6V (pg:5)

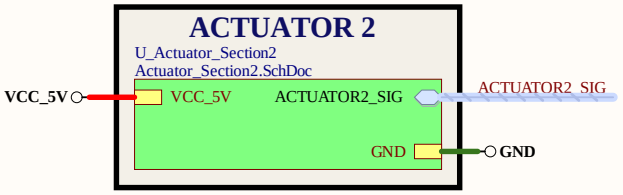
POWER NETS

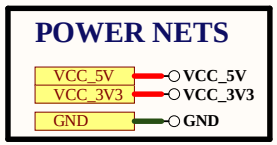


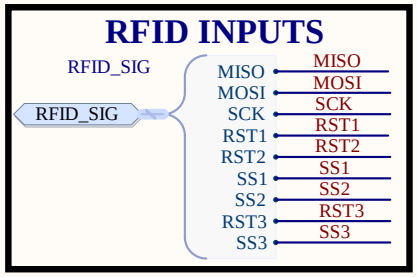

RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG
RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG


SENSOR BLOCK
U_Sensor_Section
Sensor_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
SENSOR_SIG
RFID_SIG
GND
SENSOR_SIG
RFID_SIG
GND

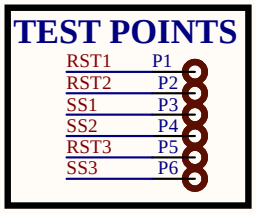

ACTUATOR 1
U_Actuator_Section
Actuator_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
ACTUATOR1_SIG
GND
ACTUATOR1_SIG
GND

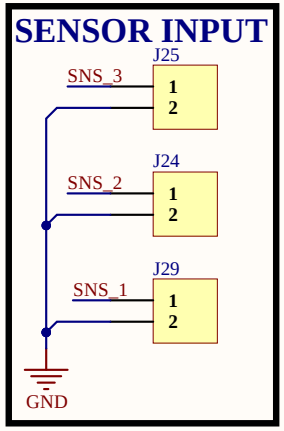

ACTUATOR 2
U_Actuator_Section2
Actuator_Section2.SchDoc
VCC_5V
VCC_5V
ACTUATOR2_SIG
GND
ACTUATOR2_SIG
GND

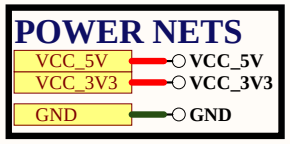

POWER NETS
VCC_5V
VCC_3V3
GND
VCC_5V
VCC_3V3
GND

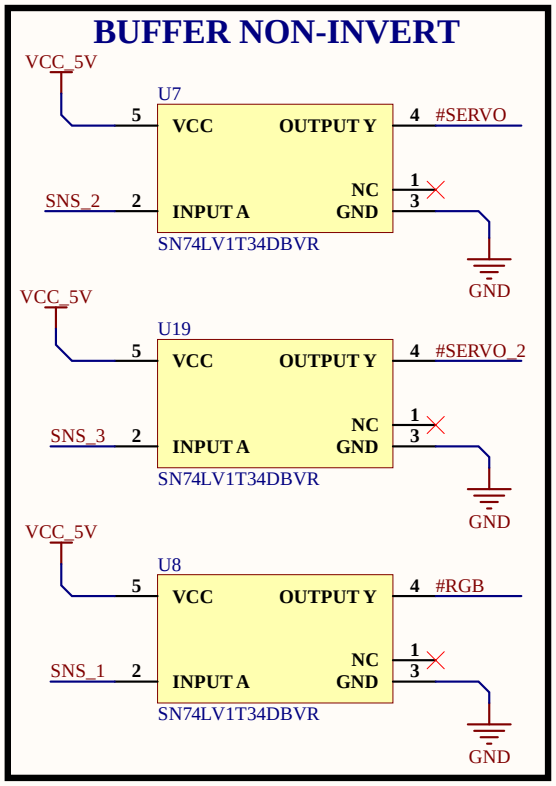


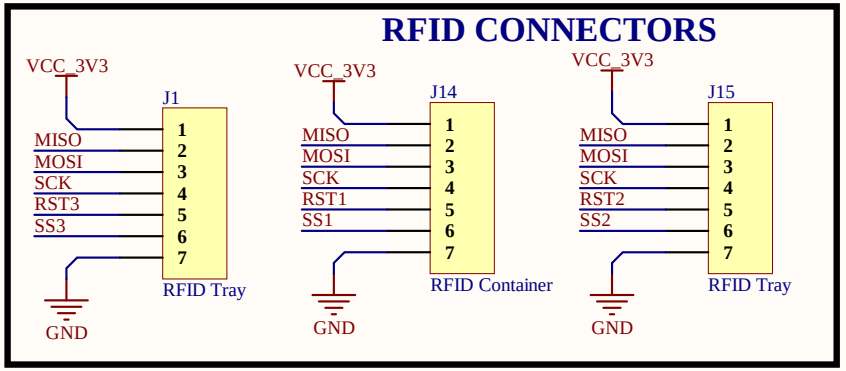




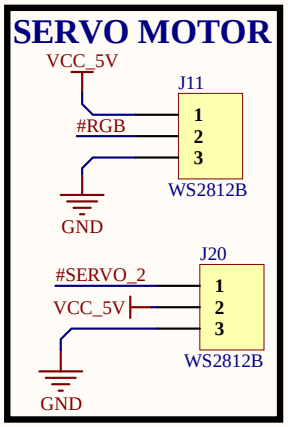


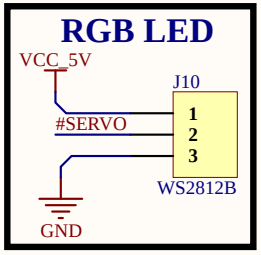


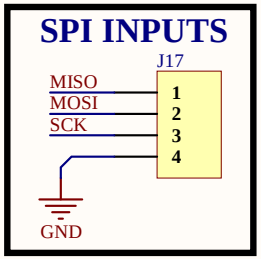




SNS signals are test signals for motor an d RGB LED.



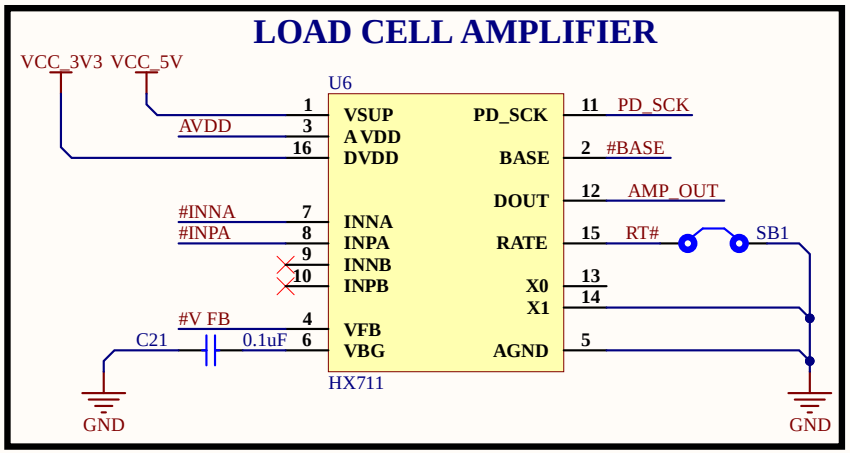




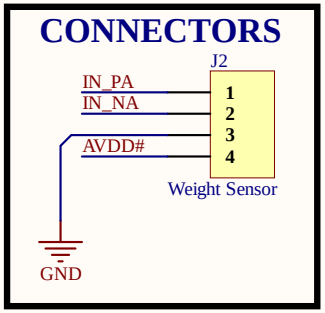
ACTUATOR1 SIGNAL



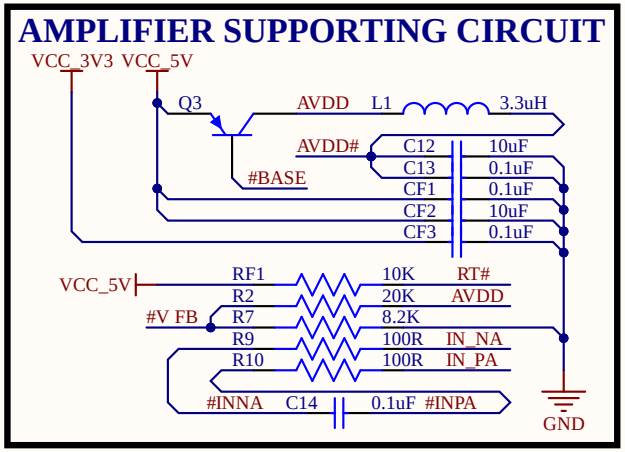
LOAD CELL AMPLIFIER



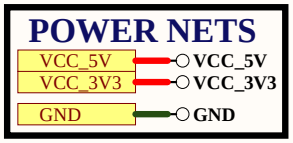
CONNECTORS



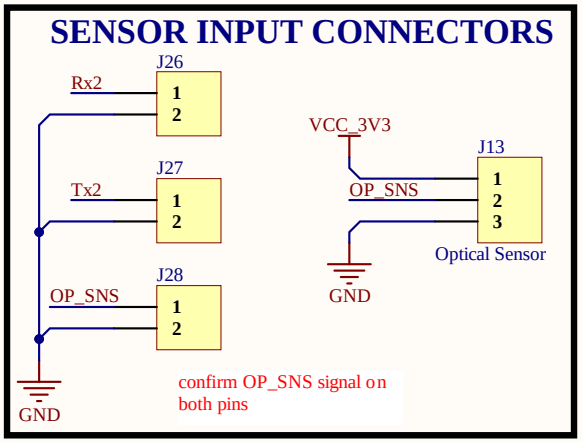
AMPLIFIER SUPPORTING CIRCUIT

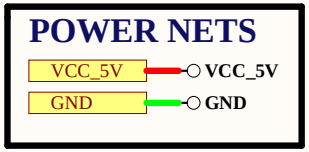


POWER NETS

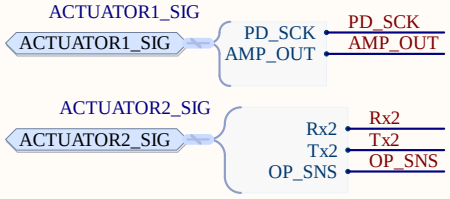



ACTUATOR2 SIGNAL
ACTUATOR2_SIG
ACTUATOR2_SIG
Rx2
Tx2
OP_SNS
Rx2
Tx2
OP_SNS


SENSOR INPUT CONNECTORS
J26
Rx2
1
2
J27
Tx2
1
2
J28
OP_SNS
1
2
GND
confirm OP_SNS signal on both pins
VCC_3V3
J13
OP_SNS
1
2
3
Optical Sensor
GND


POWER NETS
VCC_5V
VCC_5V
GND
GND

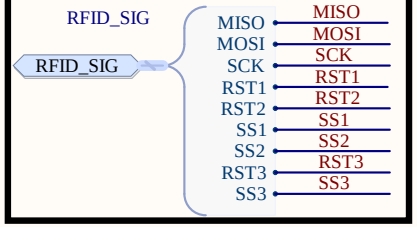
ACTUATOR SIGNALS



BUCK ENABLE



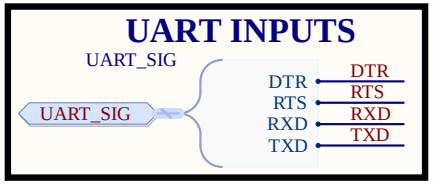
RFID INPUTS



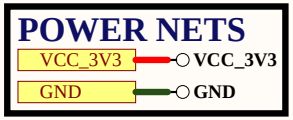
SENSOR SIGNAL



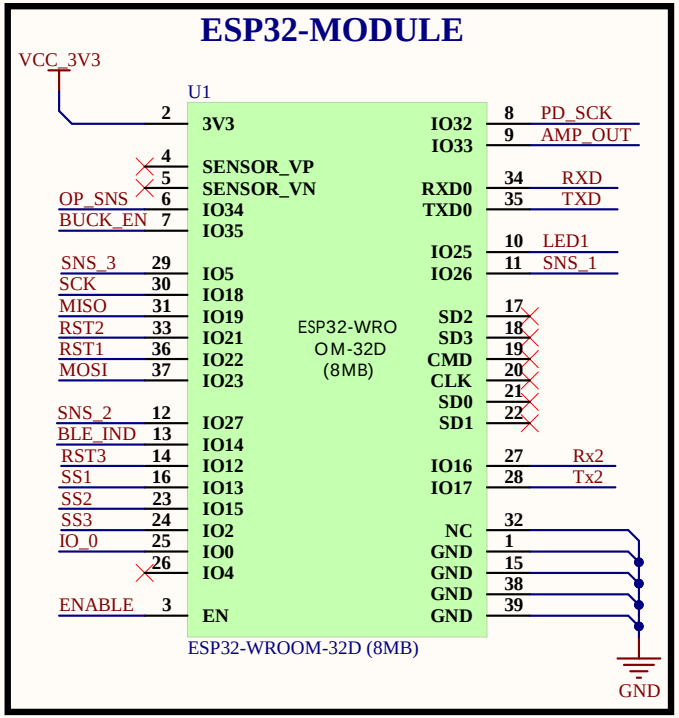
UART INPUTS



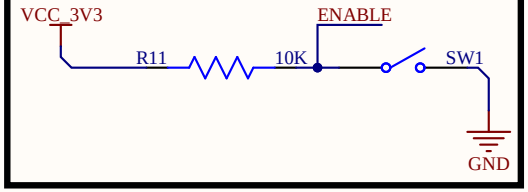
POWER NETS



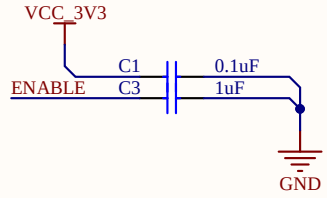
ESP32-MODULE



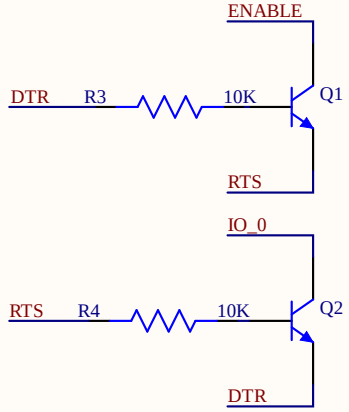
RESET SWITCH CIRCUITRY



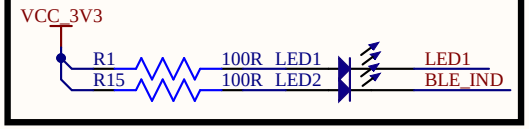
DECOUPLING CAPACITOR



SUPPORTING CIRCUITRY



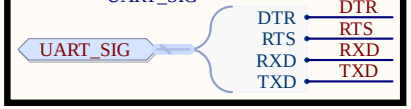
POWER INDICATION LED



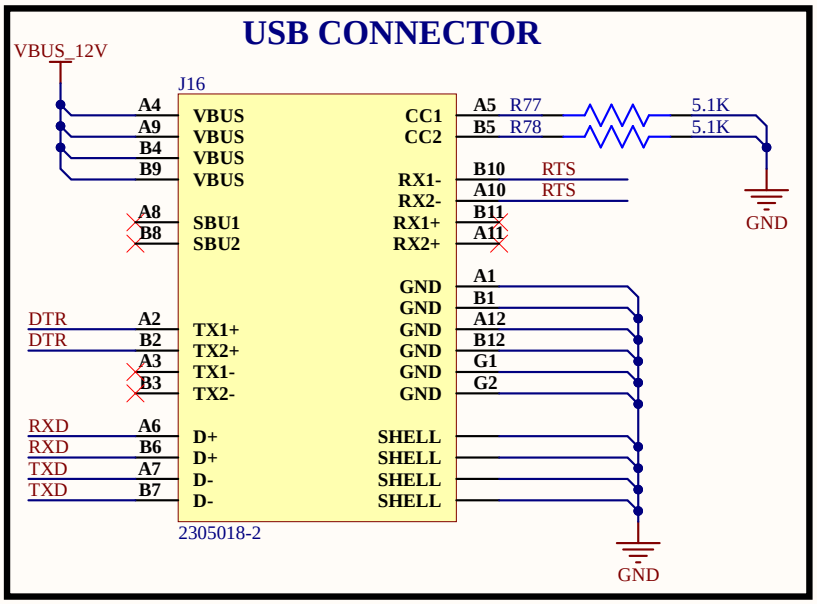
For BLue LED. forward voltage range is 2.6-3A. Taking. IF=5mA (datasheet pg: 4).
Value of current limiting RES=100R. ((3.3-2.8)/.005)

For Yellow LED, calculated resistor value is 105R.
Foward voltage: average of max and min forward voltage. IF=20mA (pg: 4)
Nearest standard value=100R

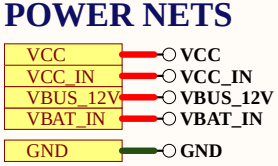
UART INPUTS



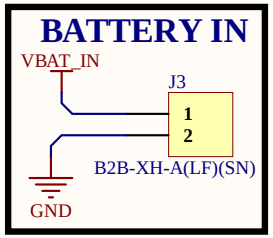
USB CONNECTOR



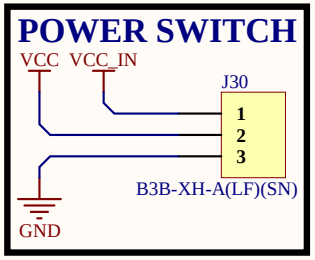
POWER NETS



BATTERY IN



POWER SWITCH



VCC and VCC_IN are externally short circuited via a jumper on the board. VCC_IN is the main power input from source selector IC.